

An Honest Momentum Backtest

How much edge survives once the future stops leaking into the past

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Abstract

This project builds a systematic cross-sectional equity momentum strategy and subjects it to a deliberately *honest* backtest. The value of the exercise is methodological rigor, not the size of any performance number: a modest-but-real net result is a success, and so is an honest null result. The headline deliverable is a “controlled demolition” that shows how much apparent edge survives each successive layer of honesty—proper point-in-time data, no-lookahead execution, realistic transaction costs, purged walk-forward validation, factor adjustment, and a multiple-testing correction. The naive equal-weighted long/short momentum strategy is found to have a *significantly negative* factor-adjusted alpha, destroyed by momentum crashes through its short leg. A gradient-boosted-tree extension that learns across six features improves raw out-of-sample performance but produces alpha statistically indistinguishable from zero: an improvement explained by known factor exposures, not genuine skill. Conceptually this is the temporal cousin of my thesis, in which an apparent spatial predictive gain vanished once spatial autocorrelation was controlled; here the analogous trap is temporal leakage.

1 Motivation: rigor over results

A backtest is the easiest thing in quantitative finance to fool yourself with. Survivorship bias, lookahead, zero-cost assumptions, k-fold cross-validation on time series, and quietly reporting only the best of many specifications each inflate performance in predictable directions. The cardinal rule of this project is therefore *rigor over results*: never optimise for a flattering Sharpe ratio, and where a choice trades correctness for a better-looking metric, surface the tension rather than make it.

The strategy under test is the classic 12–1 cross-sectional momentum signal of Jegadeesh and Titman [1]: each month, rank stocks by their cumulative return over the prior twelve months *skipping the most recent month* (to avoid short-term reversal), go long the top decile and short the bottom decile, equal-weighted and dollar-neutral. The single specification is pre-registered in a configuration file before any results are examined.

2 Data and the survivorship trap

Survivorship bias is the first and most damaging trap. The free price source (`yfinance`) only serves tickers that still trade today; building the universe from *current* S&P 500 membership and treating it as history would silently delete every delisted, bankrupt, or acquired firm and flatter returns.

We instead reconstruct **point-in-time** membership (option a). Starting from today’s constituents we walk Wikipedia’s dated change-log backwards: to recover membership as of date d we undo every change after d . The function is strictly causal—membership at d depends only on

changes dated on or before d . This recovers 340 names that have since left the index (e.g. Lehman Brothers exits 2008-08, Monsanto 2018, Yahoo 2017).

Two limitations are documented rather than hidden. First, the change-log is incomplete for older events (Bear Stearns’ removal is not recorded), so membership is treated as reliable only from 2000. Second, and more important, *the binding constraint is prices, not membership*: of 843 ever-members, `yfinance` prices 581 well, 55 partially, and 207 not at all. At the panel level 78.2% of member-months are priced and 21.8% are an unpriceable hole. Crucially, all 207 fully-missing tickers are *departed* members—the exact signature of survivorship—and the hole is worst in early history (Figure 1). The residual bias still flatters returns but far less than naive current-membership; a fully clean fix would require a paid point-in-time database such as CRSP.

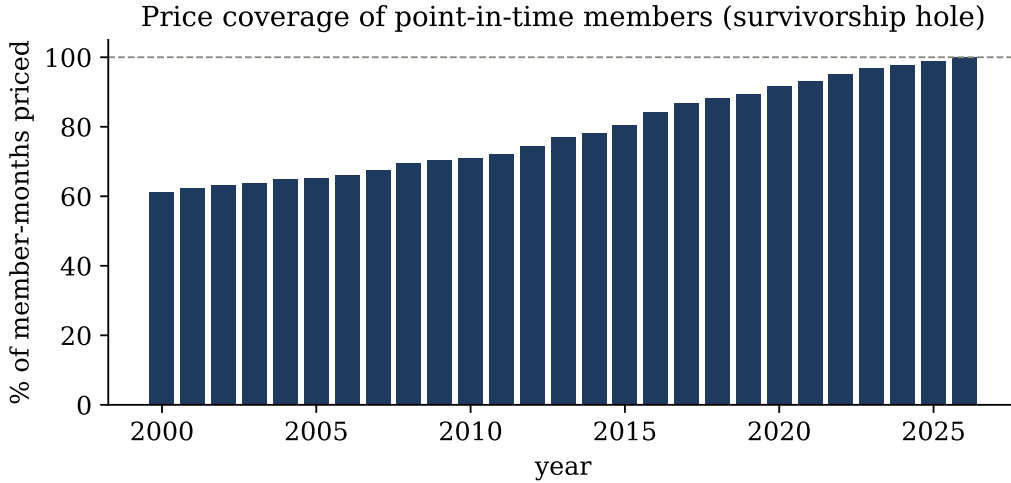


Figure 1: Fraction of point-in-time member-months for which adjusted prices are available. The shortfall is concentrated in departed names and in early history, the textbook survivorship signature. This is the irreducible residual bias of a free-data study, quantified rather than ignored.

Corrupt ticks on delisted/reused symbols (e.g. a stub priced $0.01 \rightarrow 170 \rightarrow 0.01$, implying a +3,400,000% month) are removed by a documented spike filter before any signal or return is computed: a month-end price at least $3\times$ both neighbours is nulled, with a hard cap on residual moves. This nulls 64 values and leaves genuine permanent moves untouched.

3 Signal and execution

The momentum signal at month-end t is

$$\text{signal}_t = \frac{P_{t-\text{skip}}}{P_{t-\text{skip}-\text{lookback}}} - 1, \quad \text{lookback} = 12, \text{ skip} = 1, \quad (1)$$

computed from each stock’s full price history, then masked to point-in-time membership and ranked cross-sectionally into deciles. Because row t references only prices at $t - 1$ and earlier, the contemporaneous month’s return—which forms the signal—is excluded by construction.

The backtest engine enforces the no-lookahead rule at the portfolio level: the return realised in month m uses the weights formed at $m - 1$,

$$r_m^{\text{strat}} = \sum_i W_{m-1,i} r_{m,i}, \quad (2)$$

i.e. `(W.shift(1) * R).sum()`. A mandatory unit test wires the weights to perfectly predict the *same* month’s returns; under same-bar leakage this yields an impossible Sharpe (> 8), and the test asserts the engine instead applies last month’s weights and shows no edge. A second test confirms a price spike in month t leaves signal_t unchanged but does move signal_{t+1} —the off-by-one guard.

4 Transaction costs

Round-trip cost is modelled as commission plus half-spread plus a market-impact allowance, bundled into a per-side rate. With one-way turnover $\tau_m = \frac{1}{2} \sum_i |W_{m,i} - W_{m-1,i}|$, the transacted notional is $2\tau_m$ and

$$r_m^{\text{net}} = r_m^{\text{gross}} - \frac{c_{\text{bps}}}{10^4} (2\tau_m). \quad (3)$$

Performance is reported at every level $c_{\text{bps}} \in \{0, 5, 10, 20\}$ alongside turnover ($\approx 54\%$ one-way per month), never as a single zero-cost number.

5 Validation: purged walk-forward

Plain k-fold or random splits leak the future into the past and are forbidden. We use forward-only (train-before-test) folds with two safeguards from López de Prado [6]: **purging** drops training observations whose forward-return label realises inside the test window, and an **embargo** leaves a further gap at the boundary. A unit test asserts zero label overlap between train and test across all folds. For the fixed, pre-registered momentum signal nothing is estimated on the training data, so out-of-sample performance essentially equals in-sample; the splitter becomes genuinely load-bearing for the machine-learning extension (Section 9), where labels are correlated through overlapping feature windows.

6 Factor adjustment

The real question is not whether the Sharpe is positive but whether anything survives after controlling for known factor exposure—above all momentum itself. We regress the dollar-neutral strategy’s returns on the Fama–French [2] three factors plus the momentum factor (UMD), with Newey–West (HAC) standard errors:

$$r_t^{\text{strat}} = \alpha + \beta_{\text{Mkt}} \text{MktRF}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \beta_{\text{MOM}} \text{MOM}_t + \varepsilon_t. \quad (4)$$

The baseline loads overwhelmingly on momentum ($\beta_{\text{MOM}} = +1.41$, $t = 29.6$, $R^2 = 0.81$) with a *significantly negative* alpha of -7.9% per year ($t = -3.24$); see Figure 2. The naive equal-weighted implementation is thus significantly *worse* than simply holding the well-constructed academic momentum factor.

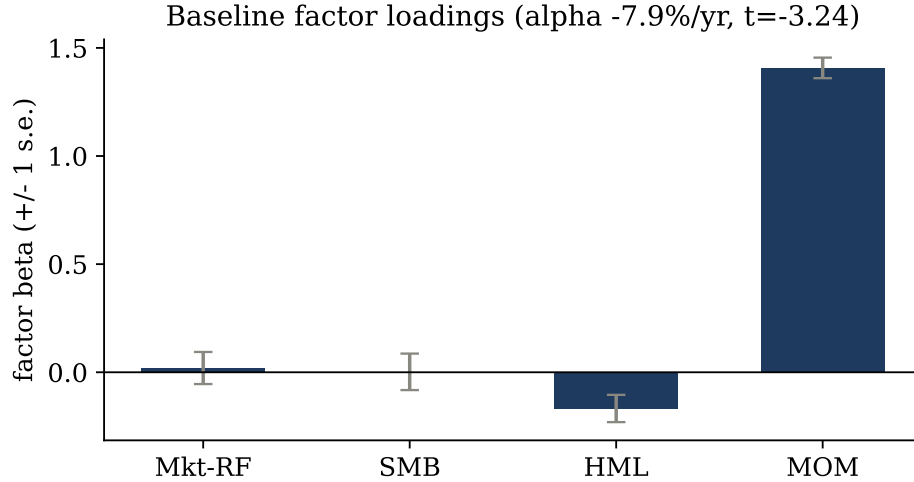


Figure 2: Baseline factor loadings (net of 10 bps), with ± 1 standard-error bars. The strategy is overwhelmingly momentum beta; the intercept (alpha) is significantly negative.

7 Multiple testing

Even a single Sharpe can look good by luck. We report the Probabilistic Sharpe Ratio and its deflated form [4], which adjust for sample length, skew, kurtosis, and—if a search were run—the expected maximum Sharpe across trials. With one pre-registered specification the deflation is minimal and the PSR against zero is 0.20: there is only a 20% probability the true Sharpe even exceeds zero. The Harvey–Liu haircut [5] would apply identically had multiple specifications been scanned.

8 The controlled demolition

Table 1 and Figures 3–4 collect the progression. Each honest layer leaves the naive strategy weaker, and it begins below water: the equal-weighted short leg is devastated by momentum crashes [3]—the worst months are 2009-03/04/05, 2002-11 and 2020-11—with a Global-Financial-Crisis sub-period Sharpe of -1.24 . Nothing survives. Under “rigor over results” this honest null/negative result is a successful outcome.

Layer	Result	Notes
1. Gross, in-sample	Sharpe -0.11	naive equal-wt L/S, zero cost
2. Walk-forward OOS	Sharpe -0.14	purged + embargoed windows
3. Net of costs (10 bps)	Sharpe -0.19	$-7.8\%/yr$ after turnover
4. Factor-adjusted alpha	$-7.9\%/yr$ ($t = -3.24$)	vs FF3+MOM; $\beta_{MOM} = 1.41$
5. Multiple-testing (PSR)	0.20	$P(\text{true Sharpe} > 0)$

Table 1: The controlled demolition of the naive 12–1 long/short momentum strategy. Each layer of honesty is applied in turn.

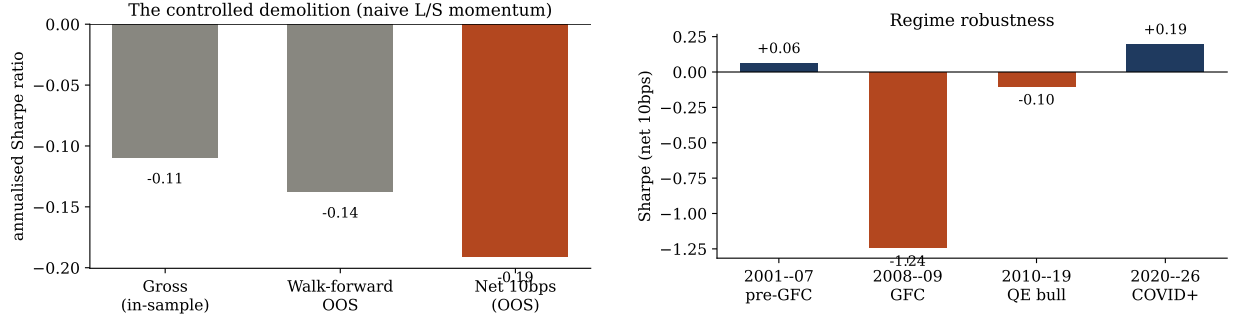


Figure 3: Left: the Sharpe ratio under successive honesty layers. Right: sub-period robustness, dominated by the Global-Financial-Crisis momentum crash.

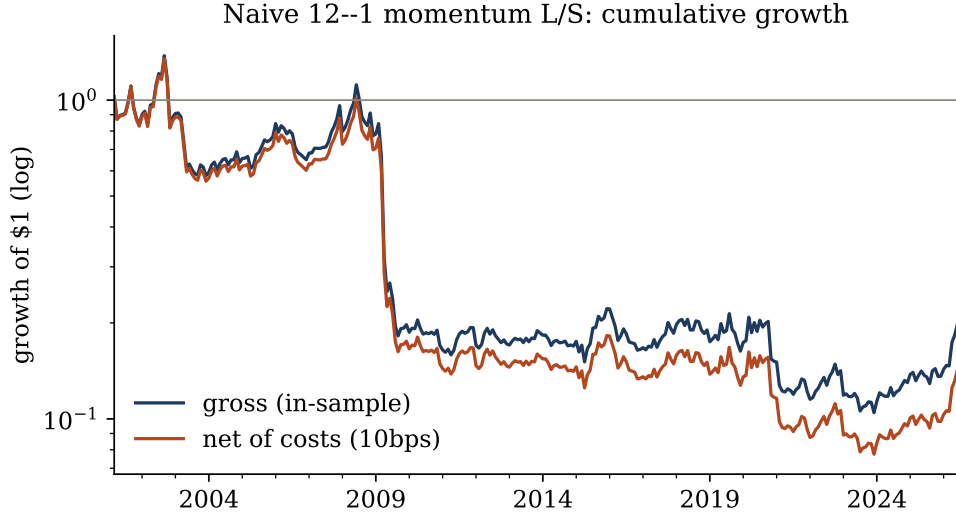


Figure 4: Cumulative growth of \$1 in the naive strategy, gross and net of costs (log scale). The drawdown through 2009 is the momentum crash.

9 Machine-learning extension

Once the ranked-signal baseline was validated, the single signal was replaced by a gradient-boosted-tree model predicting next-month cross-sectional returns from six lagged, cross-sectionally normalised features: 12–1 and 6–1 momentum, short-term reversal, trailing volatility, a long-term-reversal value proxy, and log dollar-volume liquidity (the value proxy is price-based because free data lacks point-in-time fundamentals). Predictions are generated strictly out-of-sample through the purged, embargoed walk-forward splitter, ranked into deciles, and run through the *same* engine and cost model. Hyperparameters are fixed a priori (no tuning), so there is no hidden in-sample optimisation, and the model is logged as a second pre-registered specification.¹

¹The specification names XGBoost; this environment lacks the OpenMP runtime its macOS build requires, so scikit-learn’s `HistGradientBoostingRegressor`—the same gradient-boosted-tree algorithm—is used. The methodology is identical.

Metric (2010–2026 OOS)	ML (GBT)	Naive baseline
Gross Sharpe (OOS)	0.35	0.05
Net 10 bps Sharpe	0.16	0.05
Net 10 bps ann. return	+1.2%	−0.9%
Factor-adjusted alpha	+0.6%/yr ($t = 0.27$)	—
PSR (net 10 bps)	0.75	0.58

Table 2: Machine-learning model versus the naive baseline over the same out-of-sample window. The ML model wins on raw and net Sharpe, but its factor-adjusted alpha is statistically indistinguishable from zero.

The model is a genuine improvement on raw and net performance (Table 2, Figure 5), but the decisive layer is unforgiving: its factor-adjusted alpha is +0.6% per year with $t = 0.27$, statistically indistinguishable from zero. SHAP attribution (Figure 6) explains why and confirms economic plausibility: trailing volatility (low-volatility effect), long-term and short-term reversal rank *above* the momentum features, and the model carries a *negative* momentum beta (-0.47 , $t = -5.2$). It has learned to tilt away from naive momentum and toward low-volatility and reversal, sidestepping the crashes that sank the baseline—but these are known premia, so no surviving alpha. The honest verdict is “better, but no proven edge”.

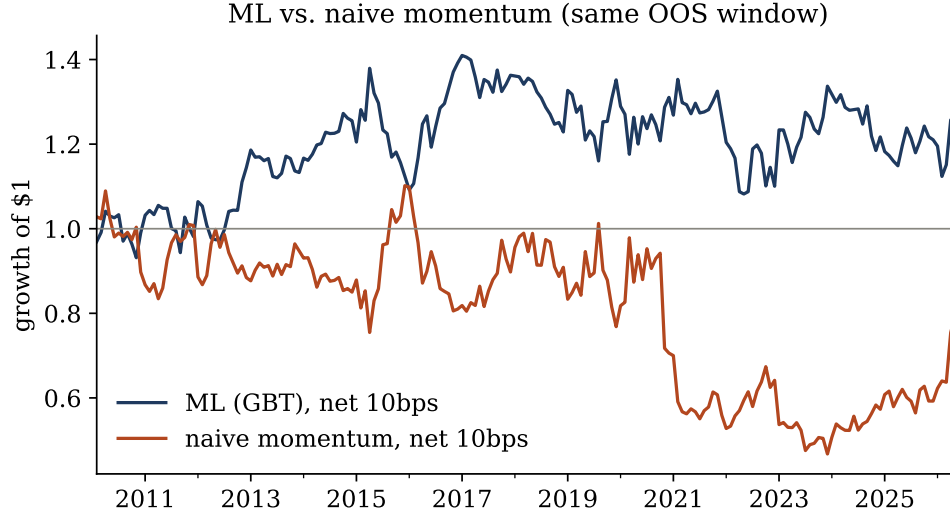


Figure 5: Cumulative net (10 bps) growth of the ML model versus the naive momentum strategy over the identical out-of-sample window.

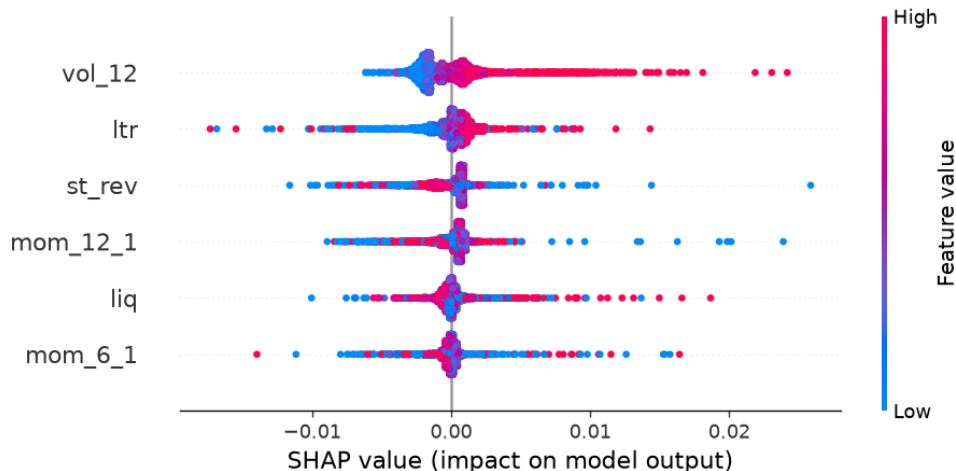


Figure 6: SHAP summary for the gradient-boosted model. Volatility and reversal features dominate the momentum features; the model effectively tilts against naive momentum, consistent with its negative momentum beta.

10 Conclusion

Two strategies, one honest conclusion. Naive equal-weighted long/short momentum does not merely fail to beat its costs; it carries a significantly negative factor-adjusted alpha, wrecked by the short leg in momentum crashes. A multi-feature gradient-boosted model improves raw out-of-sample performance, but its gains are explained by low-volatility and reversal exposure rather than genuine alpha. The deliverable is not a profitable strategy but a defensible methodology: every number is reported next to the assumption that produced it, and the apparent edge is shown to dissolve under scrutiny—exactly as an apparent spatial signal dissolved once spatial autocorrelation was controlled in the analogous thesis work.

Reproducibility. The full pipeline runs from a clean checkout with a single command (`python run.py`); all random seeds are fixed and logged, all parameters live in one configuration file, downloads are cached, and a suite of 47 tests—including the mandatory lookahead, cost, splitter, and no-leakage guards—must pass before any result is reported.

References

- [1] N. Jegadeesh and S. Titman (1993). Returns to Buying Winners and Selling Losers. *Journal of Finance* 48(1).
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- [4] D. Bailey and M. López de Prado (2014). The Deflated Sharpe Ratio. *Journal of Portfolio Management* 40(5).

- [5] C. Harvey and Y. Liu (2015). Backtesting. *Journal of Portfolio Management* 42(1).
- [6] M. López de Prado (2018). *Advances in Financial Machine Learning*. Wiley.